



NUS Orbital 2023
Milestone 3

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LifeQuest

1. Introduction	3
Team Name	3
Proposed Level of Achievement	3
Aim	3
Our Motivation	4
Our Vision	4
2. Our Posters	5
3. User Stories	9
Core User Stories	9
Extensions	9
4. Features of the System	10
User Account Authentication	10
Upgrading of Character	11
Progress Tracker	14
Adding/Removing Features	16
Player Battle System	20
5. Overall Navigation Flow	27
6. Quality Control	28
Automated Testing	28
Unit Testing	28
Integration Test	30
User Testing	31
7. Timeline and Development Plan	33

LifeQuest

1. Introduction

Team Name

StudyJio

Proposed Level of Achievement

Apollo 11

Aim

LifeQuest aims to be the ultimate gamified productivity app, turning daily routines into an adventure-filled path towards achieving goals and living a more fulfilling life.

LifeQuest

Our Motivation

Our motivation for developing this app stems from a personal desire to improve our own daily routine and establish better habits. We believe that many people struggle with finding the motivation and discipline to stick to their goals, and that a gamified approach can provide the necessary incentives to make positive changes.

Our personal experiences with these issues have inspired us to create an app that offers a fun, engaging, and rewarding way to tackle them. By combining game mechanics with productivity tools, our app will help users set goals, track progress, and stay focused on what matters most. We hope that by providing a comprehensive approach to productivity, our app can help users achieve their full potential and lead more fulfilling lives.

Our Vision

Our vision for this project is to create a leading gamified productivity app that empowers individuals to achieve their goals and live their best lives. We believe that by harnessing the power of game mechanics and social support, we can create an app that is both fun and effective in improving productivity and overall well-being.

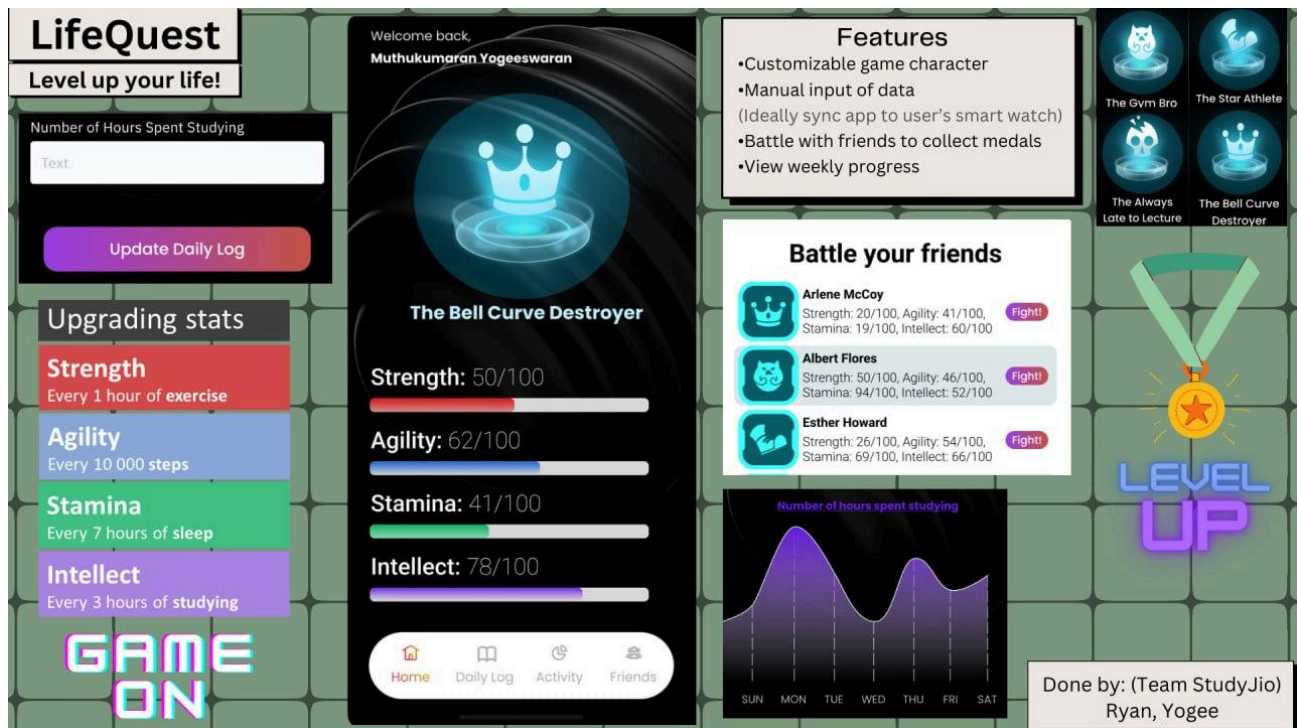
Our app will offer a variety of features, focusing on key areas such as focus and self-improvement, we aim to provide a comprehensive solution to productivity that is both practical and engaging. We envision a future where our app is the go-to solution for individuals who want to become more productive and efficient in their daily lives. By continuously improving and updating our app based on user feedback and the latest research, we hope to stay at the forefront of the gamified productivity market and make a meaningful impact on people's lives.

LifeQuest

2. Our Posters

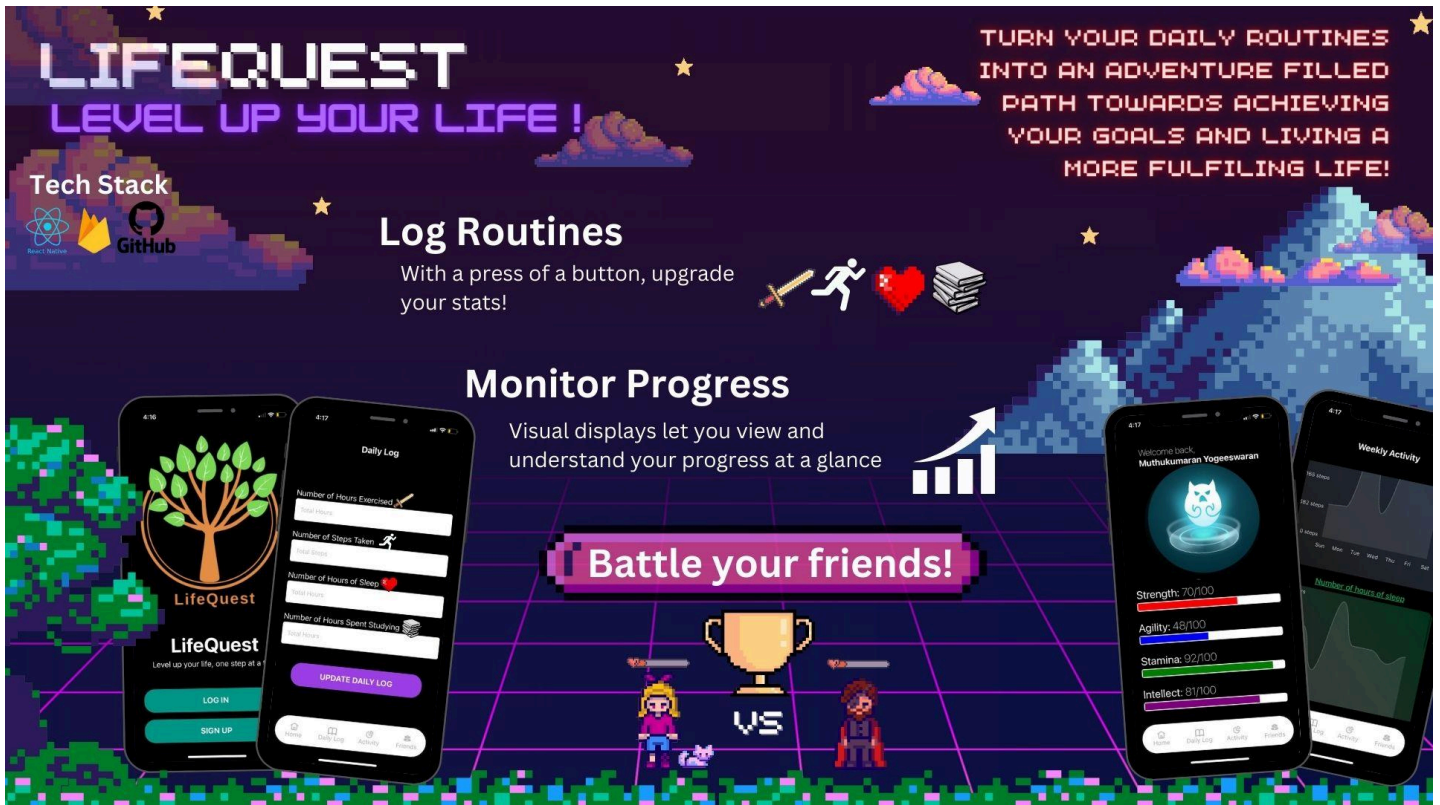
Tech Stack:

- **React Native:** For app development & front-end purposes
- **Firebase:** For back-end purposes
- **Git & GitHub:** For repository and source-code control



Lift-off poster

LifeQuest



Milestone 1 poster

LifeQuest

LIFEQUEST

LEVEL UP YOUR LIFE !

TECH STACK

React Native, Firebase, GitHub

ABOUT

LifeQuest aims to be the **ultimate gamified productivity app**, turning daily routines into an adventure-filled path towards achieving goals and living a more fulfilling life.

FEATURES

- User Account Authentication
- Upgrading of Character
- Progress tracker
- Friends system
- Player Battle System

UPGRADING STATS

STRENGTH: EVERY HOUR OF EXERCISE

AGILITY: EVERY 10 000 STEPS

STAMINA: EVERY 7 HOURS OF SLEEP

INTELLECT: EVERY 3 HOURS OF STUDYING

BATTLE SYSTEM

- Turn based automated battle system with friend bot
- Help screen available when needed
- Information box that displays moves played

GAMEPLAY!

STRENGTH: ATTACK DAMAGE

AGILITY: COMING SOON! (MILESTONE 3)

STAMINA: MAXIMUM STARTING HEALTH

INTELLECT: MAXIMUM BOOST OF ULTIMATE

SWE PRACTICES

- Version Control
 - Use of both Branching and Pull Requests while working on different systems
- Development Process
 - Utilised collaboration tools to divide and manage workloads
- Testing
 - System testing to identify edge cases and address them

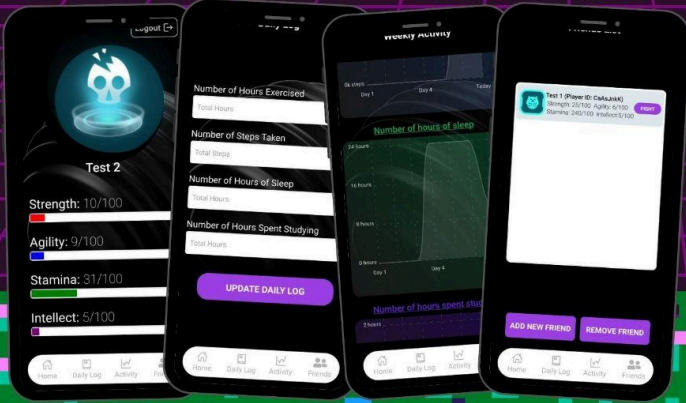
FUTURE DEVELOPMENTS

- Animation window to visualize the moves made by the players
- Character selection screen to choose avatars
- Auto tracking of sleep, steps and exercise through smart watches or smart phones

TRY OUR APP!

Username: test2@gmail.com
Password: password





Milestone 2 poster

LIFEQUEST

LEVEL UP YOUR LIFE !

TECH STACK

React Native Firebase GitHub

ABOUT

LifeQuest aims to be the ultimate gamified productivity app, turning daily routines into an adventure-filled path towards achieving goals and living a more fulfilling life.

FEATURES

- User Account Authentication
- Upgrading of Character
- Progress tracker
- Friends system
- Player Battle System
- Animation window to visualize the moves made by the players

UPGRADING STATS

STRENGTH: EVERY HOUR OF EXERCISE
AGILITY: EVERY 10 000 STEPS
STAMINA: EVERY 7 HOURS OF SLEEP
INTELLECT: EVERY 3 HOURS OF STUDYING

BATTLE SYSTEM

- Turn based automated battle system with friend bot
- Mini game to charge Ultimate!
- Help screen available when needed
- Information box that displays moves played

GAMEPLAY!

STRENGTH: ATTACK DAMAGE
AGILITY: CHANCES OF DODGING AN ATTACK!
STAMINA: MAXIMUM STARTING HEALTH
INTELLECT: MAXIMUM CHARGE OF ULTIMATE!

SWE PRACTICES

- Version Control
 - Use of both Branching and Pull Requests while working on different systems
- Development Process
 - Utilised collaboration tools to divide and manage workloads
- Testing
 - System testing to identify edge cases and address them

FUTURE DEVELOPMENTS

- Character selection screen to choose avatars
- Auto tracking of sleep, steps and exercise through smart watches or smart phones

DOWNLOAD EXPO & TRY OUR APP!

Username: test2@gmail.com
Password: password



3. User Stories

Core User Stories

- 1) As a student, I want to embrace a healthy lifestyle that prioritises sleep, exercise and studying, allowing me to balance my responsibilities without sacrificing my well-being.
- 2) As a student who lacks enthusiasm for studying, I want to be able to make completing my assignments feel more rewarding and satisfying.
- 3) As a student, I want to be able to keep track of my progress to ensure I am not neglecting any important elements of my personal growth and development.
- 4) As an unmotivated individual, I want some healthy competition that drives me to put more effort into being productive and living a balanced lifestyle.

Extensions

- 1) As an individual, who lacks the motivation to use tracking apps over a long period of time, I want to be incentivised and encouraged to use the app.
- 2) As a player in the game, I want to be able to personalise and customise my character/profile to enhance the overall gaming/user experience.
- 3) As a user who enjoys social interaction, I want to be included in a community where I can connect with like-minded individuals, share progress, and provide support and encouragement.
- 4) As a long-term user, I want a convenient app that can seamlessly integrate with smart devices, collecting data effortlessly as I go about my daily life.
- 5) As a player, seeking continuous engagement, I want to be regularly introduced to new challenges, quests, or activities to keep me motivated and invested in the app.

4. Features of the System

User Account Authentication

Description:

As LifeQuest is a personalised productivity app, each user will need to be authenticated with his/her own unique account. Currently, LifeQuest has implemented 1 main sign-in method, which allows users to create an account using their email and password.

Utilising the Firebase SDK Authentication, users can:

- Login via email and password
- Create an account if the user does not yet have an existing account
- Retrieve/Reset a forgotten password (a password reset email will be sent to their registered email)

Based on the user's action, the relevant call to the Firebase authentication instance is made. For instance, registering with an email and password will call `createUserWithEmailAndPassword`, whereas resetting an account's password will call `sendPasswordResetEmail`.

Only upon successful login or account registration will the user be routed to the next page, and the account reflected in the Firebase authentication.

Implementation philosophy:

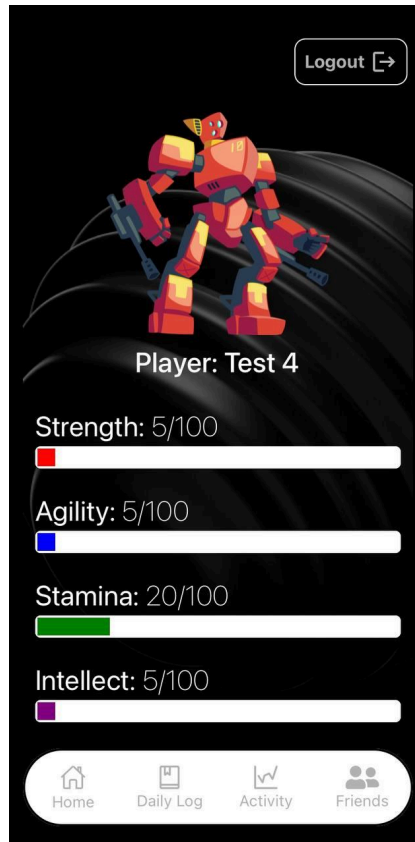
Within the authentication controller, a listener constantly checks if the user is signed in or not. When the user first opens the mobile app, the authentication controller will first check if the user is logged in. If the user is logged in, he/she will be able to access the Home page.

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Upgrading of Character

Description:

After new users have successfully signed up, they will be given an avatar that can be upgraded and used to battle their friends. All players will be given an avatar with the same base stats. How these character stats come into play will be further elaborated below



In order to upgrade these 4 key character stats, players will be required to carry out specific tasks:

1. Strength: Every hour of exercise will level up the character's strength by 1 point
2. Agility: Every 10,000 steps walked will level up the character's agility by 1 point
3. Stamina: Every 7 hours of sleep taken will level up the character's stamina by 1 point
4. Intellect: Every 3 hours spent studying will level up the character's intellect by 1 point

Users will have to manually key in their daily progress, which will be reflected in the progress bar of their character's respective traits.

LifeQuest

Implementation philosophy:

The user's progress will be stored within our Firestore Database. When users submit their daily log, values such as "total_exercise" and "total_study" will be incremented by the respective hours inputted. For the number of steps taken, the value will be rounded and stored to the nearest thousand (E.g 12812 steps will be taken as 13k steps)

To ensure that our character's progress bar is updated in real-time based on user input, we made use of a listener that calls these values from the Firestore Database whenever the Home page is rendered.

Submitting the daily log:

Daily Log

Number of Hours Exercised
10

Number of Steps Taken
12812

Number of Hours of Sleep
7

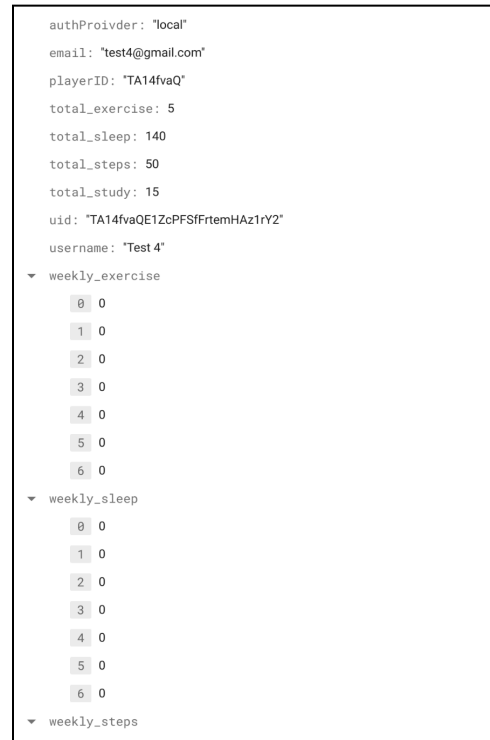
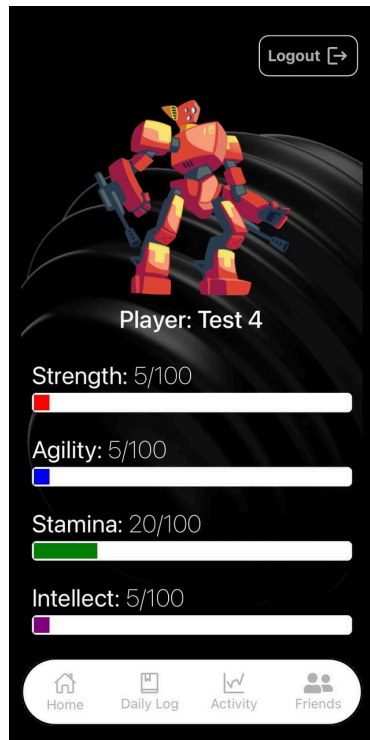
Number of Hours Spent Studying
6

UPDATE DAILY LOG

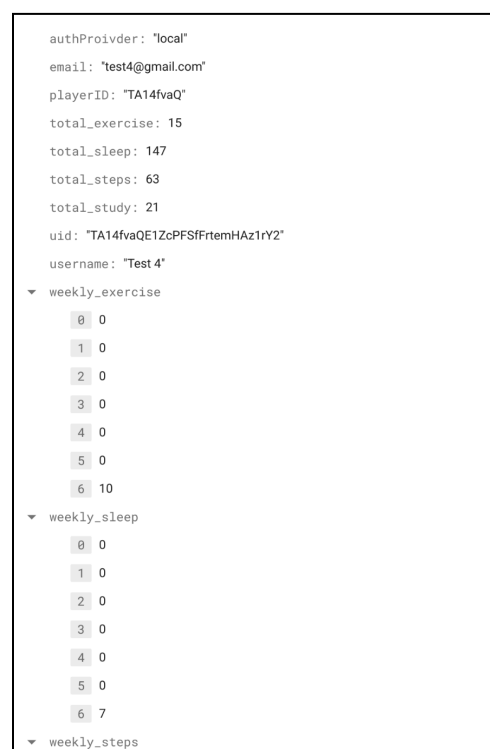
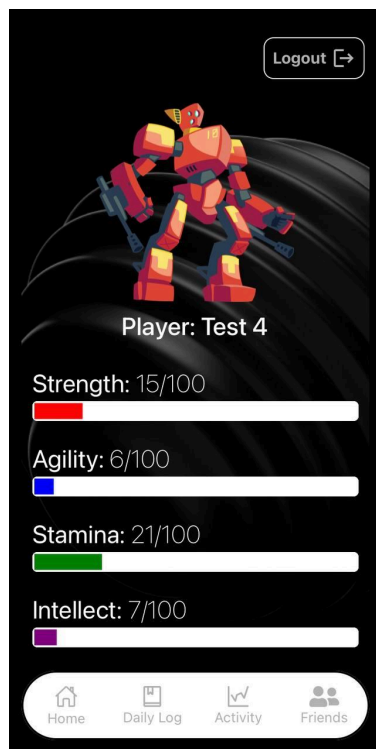
Home Daily Log Activity Friends

LifeQuest

Before submitting the daily log:



After submitting the daily log:

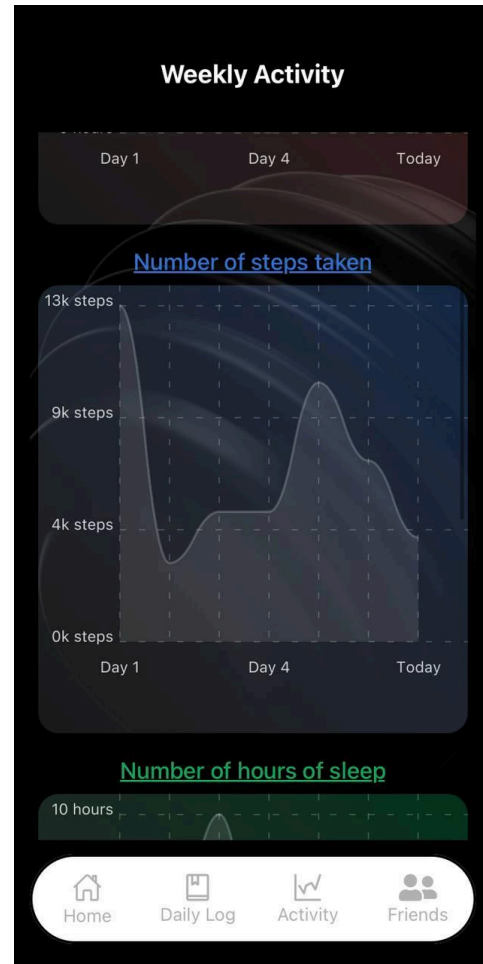
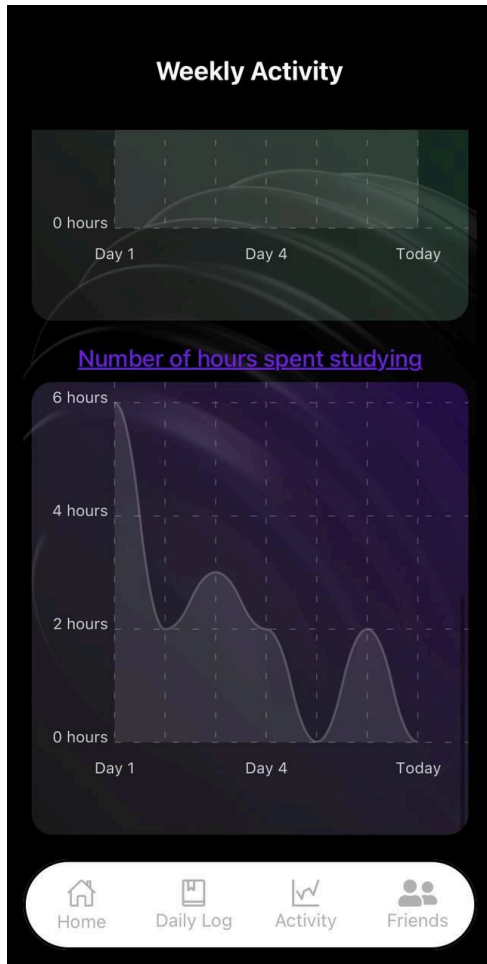


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Progress Tracker

Description:

Users will be able to visualise how their daily habits change over time through the use of a line graph. 4 line graphs will be displayed, each displaying the changes in the user's respective habits over the last 7 days – number of hours exercised, number of steps walked, number of hours of sleep received, and number of hours spent studying.



LifeQuest

Implementation philosophy:

A weekly log feature that tracks and stores a player's exercise hours for each day of the week. Within the database, we have a list structure that represents the past 7 days, with each element in the list representing the data for a specific day.

When a user submits their daily log, the least recent value is removed from the list, and the new input provided by the user is then added to the list. This process ensures that the weekly log always reflects the player's hours/steps for the most recent 7 days. The Firestore database's real-time capabilities ensure that the log is seamlessly updated across devices, allowing users to view their updated progress instantly.

Daily Log

Number of Hours Exercised
6

Number of Steps Taken
6000

Number of Hours of Sleep
6

Number of Hours Spent Studying
6

UPDATE DAILY LOG

Home Daily Log Activity Friends

Before Daily Log is submitted:

▼ weekly_exercise
0 7
1 1
2 0
3 3
4 2
5 0
6 6
▼ weekly_sleep
0 6
1 10
2 4
3 7
4 7
5 6
6 6
▼ weekly_steps

After Daily Log is submitted:

▼ weekly_exercise
0 10
1 7
2 1
3 0
4 3
5 2
6 0
▼ weekly_sleep
0 7
1 6
2 10
3 4
4 7
5 7
6 6
▼ weekly_steps

LifeQuest

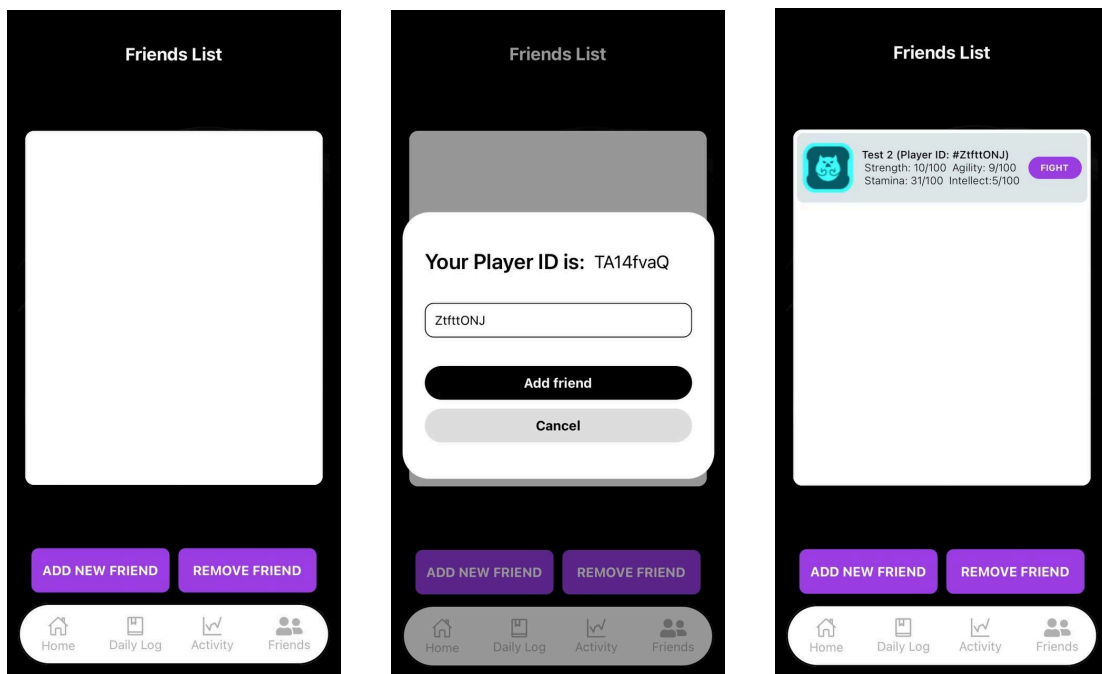
Adding/Removing Features

Description:

LifeQuest's add/remove friend feature is designed to enhance social interaction and collaboration within the app. Once a user has created an account and signed in, they can utilise this feature to connect with other users and establish a network of friends within the LifeQuest community.

To add a friend, the user can search for other registered users within the app by their player ID. Upon finding the desired user, they can send a friend request, and both users will be added to each other's friend lists. This allows them to view each other's profiles, track each other's progress, and battle each other's avatar. Conversely, the remove friend feature enables users to disconnect from friends they no longer wish to associate with within the app. By accessing their friend list, a user can select the friend they want to remove and initiate the removal process using their player ID. Once confirmed, the connection between the two users will be severed, and they will no longer have access to each other's profiles.

Overall, the add/remove friend feature in LifeQuest serves as a means to foster a sense of community, encourage collaboration, and facilitate social engagement among users of the app.



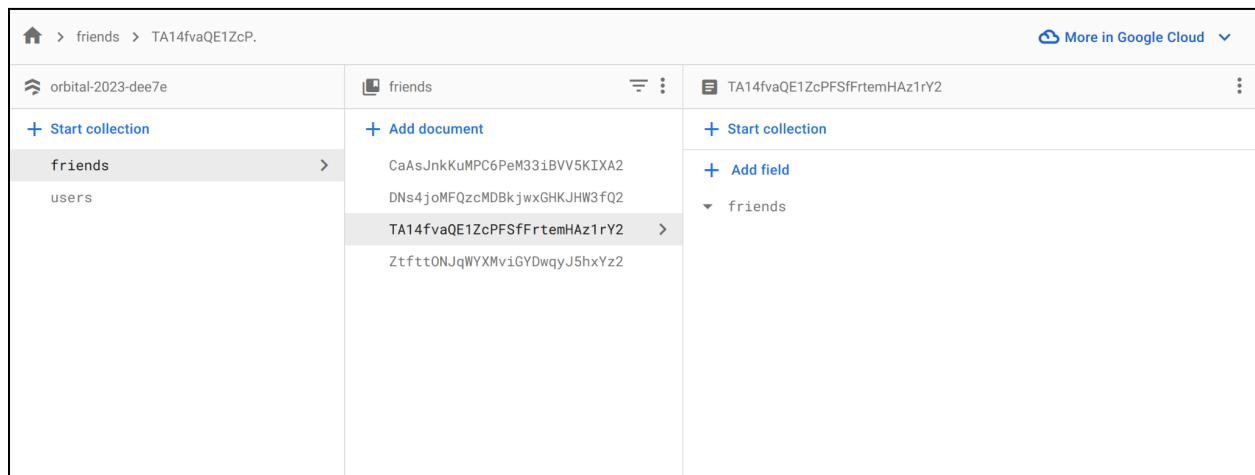
LifeQuest

Implementation philosophy:

The add/remove friend feature in our Firebase and React Native-based application allows users to connect and disconnect with other users using their player ID. Here's how the feature works:

1. **Authentication:** Users are required to create an account and sign in using Firebase Authentication for secure access to the app.
2. **Database Structure:** Our Firebase database is organised into collections. The "users" collection stores user information, including player IDs, usernames, and other relevant data. The "friends" collection manages friend connections between users.
3. **Adding/Removing Friends:** To add/remove a friend, the app iterates through the "users" collection to locate the user with the specified player ID. Once found, the user's database and data are retrieved from the document. The app then searches the "friends" collection to find the user's player ID and updates the corresponding document by adding/removing the friend's data, such as their player stats. Similarly, the app iterates through the "friends" collection to find the friend's player ID and updates the corresponding document by adding/removing the user's data, establishing a mutual connection.

By utilising Firebase's real-time capabilities, our add/remove friend feature ensures that friend connections are instantly updated across all users' devices. This enhances social interaction and facilitates seamless collaboration within our application.



LifeQuest

Both players accounts before the friend request is sent:

More in Google Cloud	
friends	TA14fvaQE1ZcPFSfFrtemHAz1rY2
+ Add document	+ Start collection
CaAsJnkKuMPC6PeM33iBVV5KIXA2	+ Add field
DNs4joMFQzcMDBkjwxGHKJHW3fQ2	▼ friends
TA14fvaQE1ZcPFSfFrtemHAz1rY2	
ZtfttONJqWYXMviGYDwqyJ5hxYz2	

More in Google Cloud	
friends	ZtfttONJqWYXMviGYDwqyJ5hxYz2
+ Add document	+ Start collection
CaAsJnkKuMPC6PeM33iBVV5KIXA2	+ Add field
DNs4joMFQzcMDBkjwxGHKJHW3fQ2	▼ friends
TA14fvaQE1ZcPFSfFrtemHAz1rY2	
ZtfttONJqWYXMviGYDwqyJ5hxYz2	

LifeQuest

Both players accounts after the friend request is sent:

More in Google Cloud

friends

+ Add document

CaAsJnkKuMPC6PeM33iBVV5KIXA2

DNs4joMFQzcMDBkjwxGHKJHW3fQ2

TA14fvaQE1ZcPFSfFrtemHAz1rY2

ZtfttONJqWYXMviGYDwqyJ5hxYz2

TA14fvaQE1ZcPFSfFrtemHAz1rY2

+ Start collection

+ Add field

friends

0

agility: 9

friendID: "ZtfttONJ"

icon: "../../assets/player_avatars/gym_bro.png"

intellect: 5

stamina: 31

strength: 10

title: "Test 2"

uid: "ZtfttONJqWYXMviGYDwqyJ5hxYz2"

More in Google Cloud

friends

+ Add document

CaAsJnkKuMPC6PeM33iBVV5KIXA2

DNs4joMFQzcMDBkjwxGHKJHW3fQ2

TA14fvaQE1ZcPFSfFrtemHAz1rY2

ZtfttONJqWYXMviGYDwqyJ5hxYz2

ZtfttONJqWYXMviGYDwqyJ5hxYz2

+ Start collection

+ Add field

friends

0

agility: 9

friendID: "TA14fvaQ"

icon: "../../assets/player_avatars/gym_bro.png"

intellect: 10

stamina: 26

strength: 28

title: "Test 4"

uid: "TA14fvaQE1ZcPFSfFrtemHAz1rY2"

LifeQuest

Player Battle System

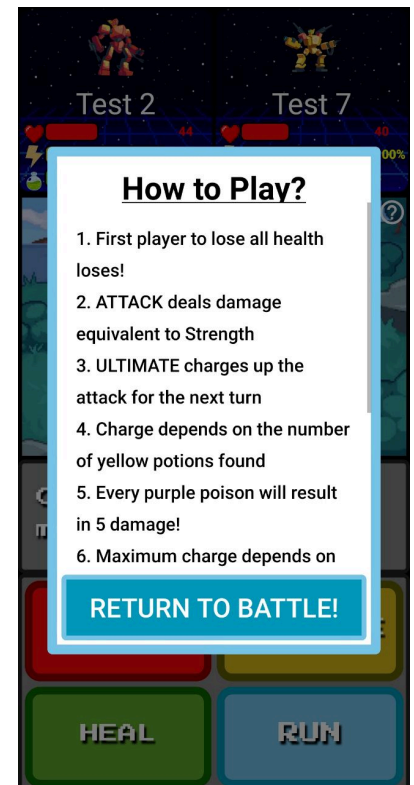
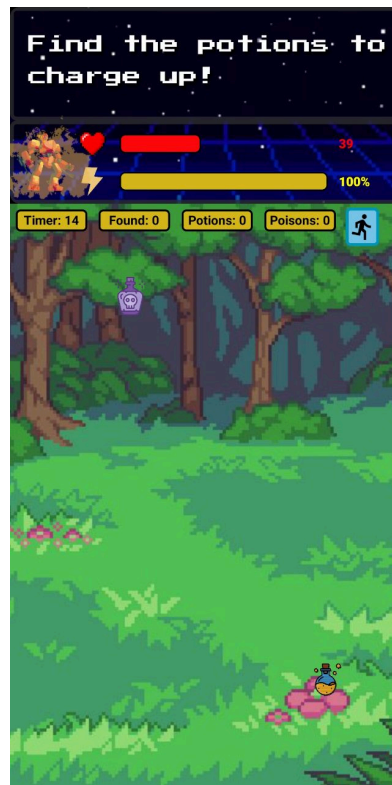
Description:

Each user will be tagged with a unique player identification number (ID). Through using these IDs, users will be able to add other players to their friend's list. Once added, the user's avatar will be allowed to battle their friend's avatar.

Outlined below is how the battle system will be implemented, as well as how the avatar's character stats come into play:

- Strength: Determines how much damage is done during each attack
- Agility: Determines the chances of dodging an attack.
- Stamina: Determines the starting health of the character where 100 is the max health for all players.
- Intellect: Determines the maximum number of potions that can be collected in the mini game (**centre picture**) that is triggered by the ultimate move. Every yellow potion collected is equivalent to charging 2 normal attack for the next turn. Every wrong purple poison collected will result in 5 damage done to the player.

In the game, the player's health bar is updated based on their character's stats, and they can restore health using healing abilities limited to 3 uses per game. The player can charge their ultimate ability as long as their power bar is not depleted. To access information about how to play, they can click the info button in the top right corner of the animation window. Once the player makes a move, the game starts, and a bot randomly chooses a move in response. The damage taken and moves played are shown in a grey box below the animation window.



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Implementation philosophy:

The battle system implements game logic using boolean values to track move actions and state changes. These boolean values enable move validation, enforce move limits, manage game state, and control the flow of the game. By utilising these boolean variables, the system determines the validity of moves, adjusts actions based on move limits, and ensures proper sequencing of player and enemy turns. Additionally, the system evaluates health conditions at every move to determine the outcome of the battle. Overall, the use of boolean values in the battle system enhances the gameplay experience by providing structured decision-making and result evaluation.

Features implemented:

In order to make the gameplay smooth and straightforward to play, we made the game intuitive and restricted moves based on different scenarios to prevent any errors from arising. We considered several test cases as well to address while implementing the features.

1. Disabled the move buttons when it's the bot's turn to prevent players from making any moves and disrupting the game

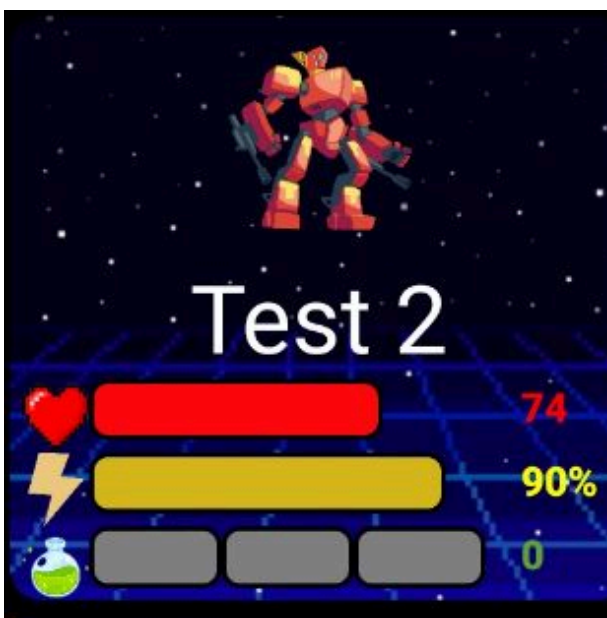


LifeQuest

2. Disabled the heal button when the player's health is at the maximum 100 health, to prevent players from wasting a heal which won't heal any health

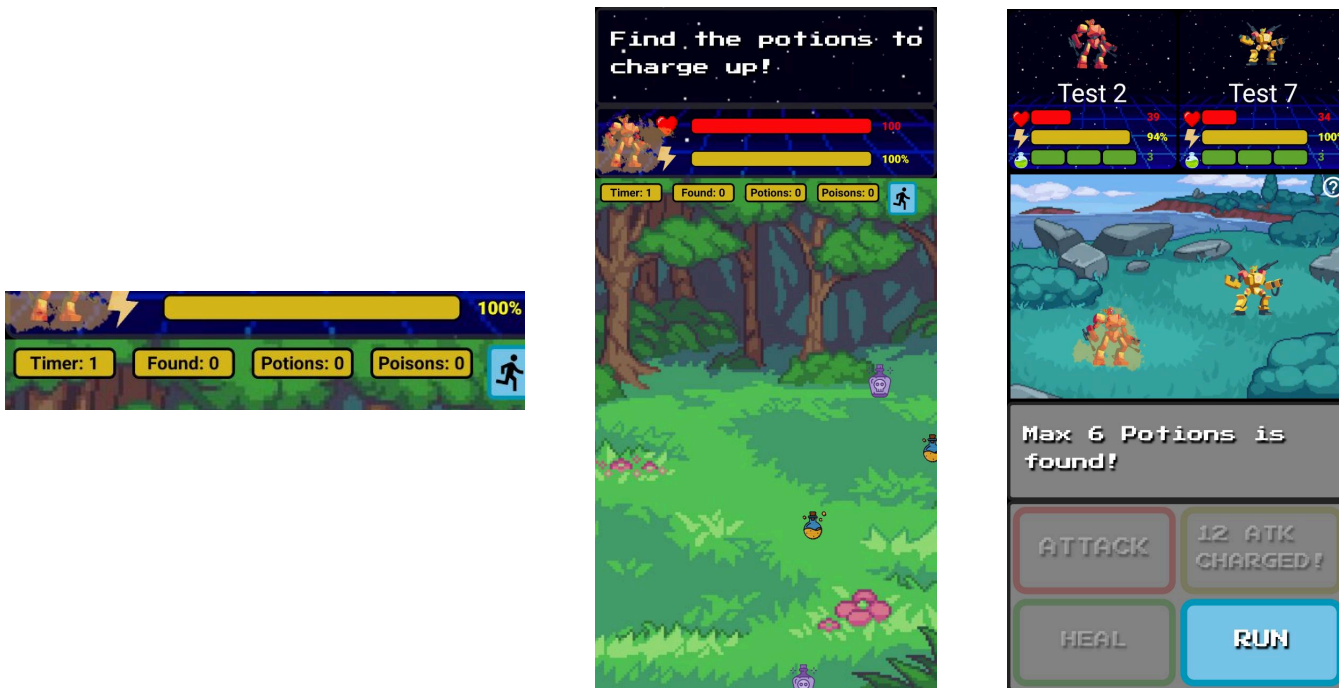


3. Disabled the ultimate and heal button when the ultimate or heal has been completely used up, in order to let the players know and plan their next move. Players will also be able to view how many heals or power they have left in the top section.

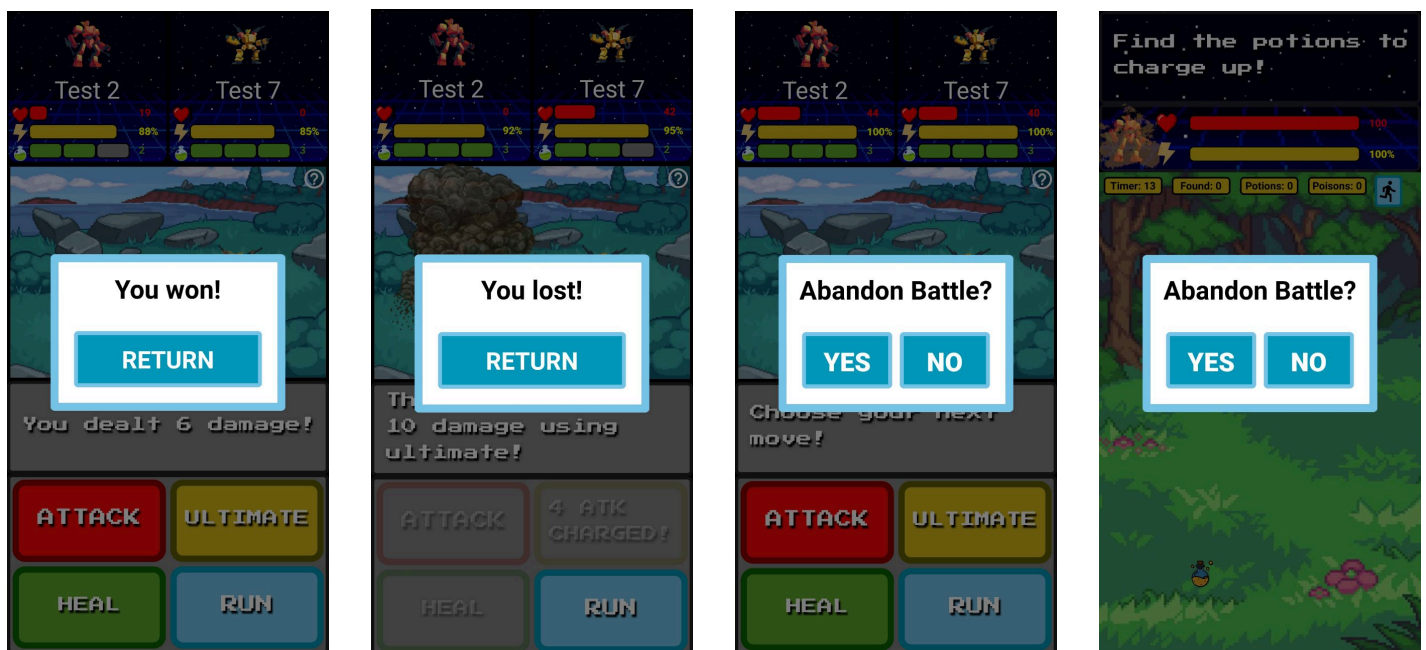


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4. Made the charging of ultimate a mini game and the game stops and closes when either the time limit of 15s has run out, power bar has been depleted or the maximum number of potions equivalent to the intellect stat has been found.



5. Alert players of the outcome of their battle or when they are about to leave the battle. Players are able to leave the battle by pressing the blue coloured RUN button or the blue coloured exit icon while in the ultimate mini game.



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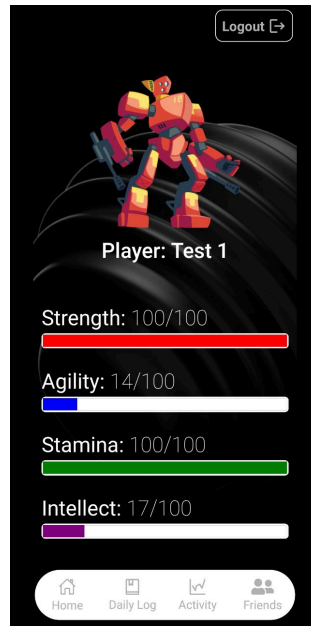
6. Introduced animations to visualise the moves played by the players.



LifeQuest

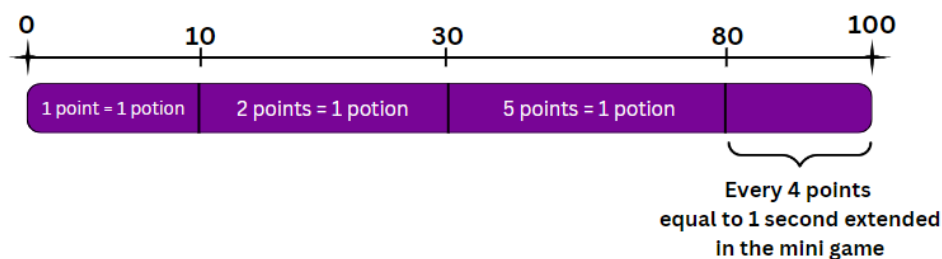
Possible Issues that could arise in the long term

1. Players maxing out their stats in the long term could make the game aspect of LifeQuest boring as players would defeat their opponents in 1 turn.



To fix this issue, in the future an experience points system could be implemented where the max for strength and agility will be capped at 30 to prevent the players from becoming too strong and dodging every attack. Any additional stat gained in strength, agility or even Stamina (after reaching max 100 health) will result in adding experience for the player instead, where these points could be used for character customization.

For the Intellect Stat which corresponds to the maximum number of potions the player could collect in the mini game, since collecting 100 potions within 15 seconds seems unrealistic. The Intellect stat could be implemented into the game such as shown below in the future.



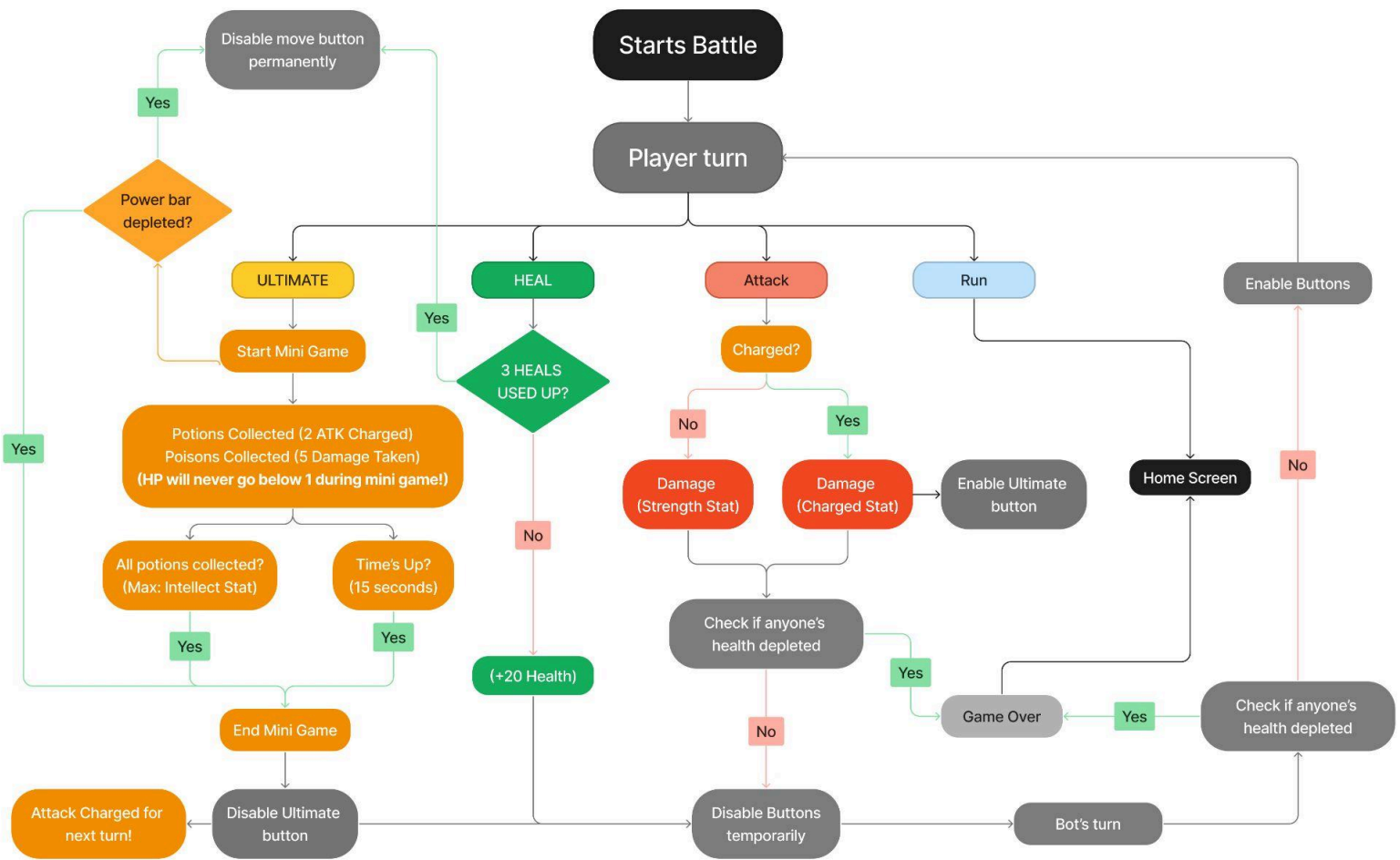
Maxed out intellect Stat

Max Potions: 30

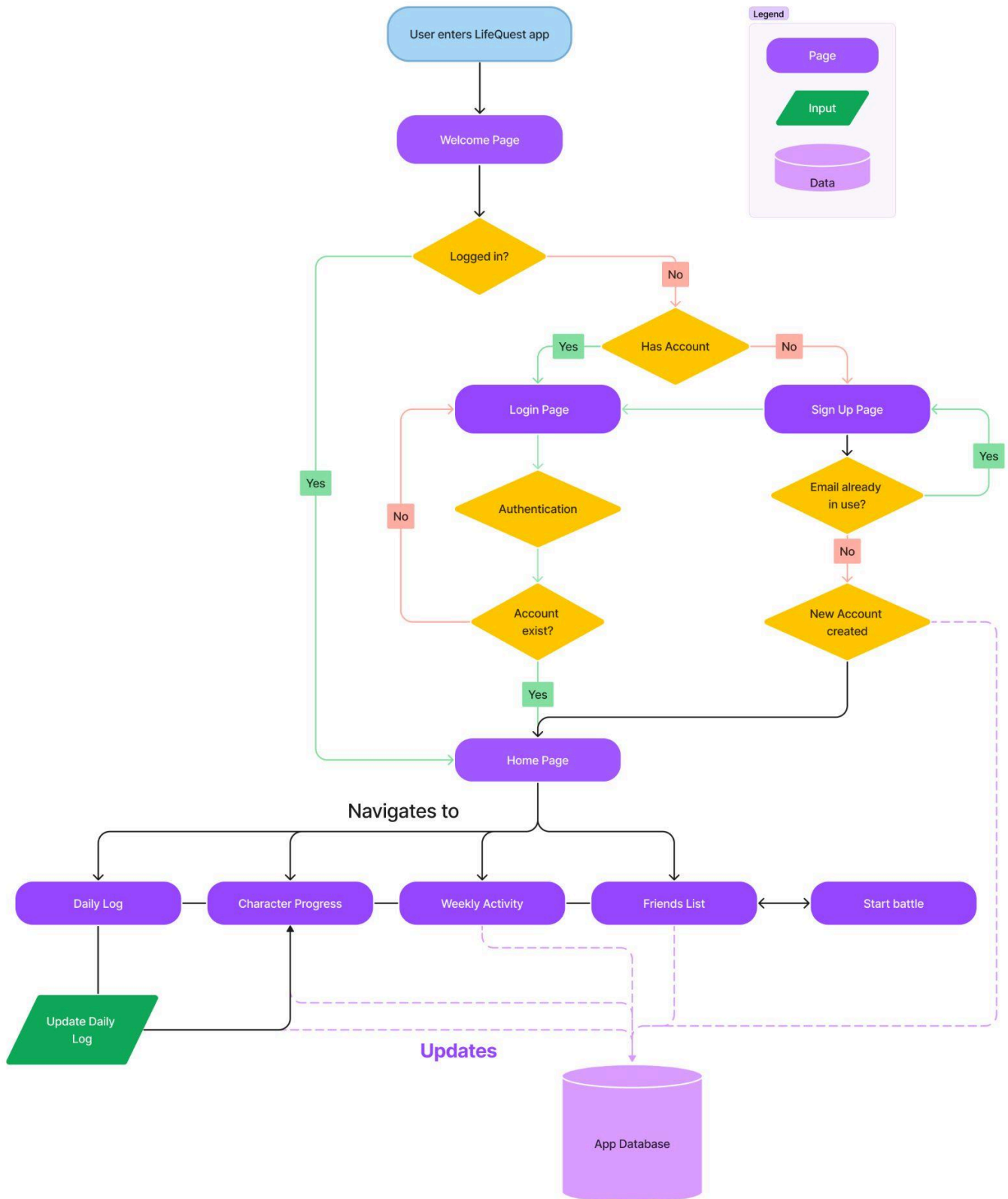
Max time: 20 seconds

LifeQuest

Diagrams:



5. Overall Navigation Flow



6. Quality Control

Automated Testing

Automated tests are essential to test features and fix bugs to ensure the app performs as expected. For LifeQuest, we used two main categories of automated testing:

1. Unit test: tests a single function, method or class
2. Integration test: tests a complete app or a certain navigation path of an app

Unit Testing

Unit testing is the part of the automated testing process that contains small units of code that test specific parts of the program for their reliability. For LifeQuest, unit tests were written and conducted to focus on testing the logic behind selected functions within each of the controllers:

- For the UI components, we ran a snapshot test to ensure that all the components were rendered correctly
- For intractable components, such as through a button click, we wrote test to simulate clicking and recorded the responses to see if the components were working as intended
- For functions with limits on the inputs, such as our “Daily Log” page, we ran boundary tests on these functions to ensure that they work as intended, and that there are no unintended outputs

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Module	Testing Objective	Expected Results	Pass / Fail
Registration and Authentication			
Email Validation	Test when user does not fill up all input fields	Expects an error message of "Please fill up all fields"	Pass
	Test when user enters an email that is already in use	Expects an error message of "Email already in use. Please enter a valid email"	Pass
	Test when user inputs a username that is longer than allowed	Expects an error message of "Username has a 12 character limit"	Pass
Username validation	Test if users are able to log into the app with an invalid email or password	Expects an error message of "Invalid email/password"	Pass
AppButton component	Test if component renders successfully	Successful	Pass
Updating Daily Progress			
Home Page	Check that the Progress bars will not exceed a limit of 100	Users are able to update their hours and steps	Pass
Daily Log Page	Test when user does not fill up all input fields	Expects an error message of "Please ensure all fields are filled correctly"	Pass
	Test when user enters invalid inputs (Example: entering a string instead of a number)	Expects an error message of "Please ensure all fields are filled correctly"	Pass

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	Test when user enters an unrealistic amount (Example: entering a cumulative of more than 24 hours/ entering more than 20k steps)	Expects an error message of "Please ensure all inputs are within 24 hours"	Pass
<i>Friends & Battle System</i>			
Friends Page	Test when user adds/removes a player ID that does not exist	Expects an error message of "Invalid player ID entered"	Pass
	Test when user adds/removes their own player ID as a friend	Expects an error message of "Cannot enter your own ID"	Pass

Integration Test

Integration testing is used to simulate and mimic user interactions with the app such as scrolling, selecting, entering certain texts and clicking on buttons.

<i>Module</i>	<i>Testing Objective</i>	<i>Expected Results</i>
Signing Up/ Logging In	Test the ability to create a new user account	Users are able to create a new account
	Test the ability to log into the app	Users are able to log into the app
Updating Daily Logs	Test the ability of the app to update daily hours and steps	Users are able to update their daily statistics

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User Testing

Apart from automated testing, user testing was conducted with personnels whose criteria meets the target audience. User testing is the process of having end users test and evaluate the product or feature.

Module	Feedback Received	Follow-Up Action	Remarks
Registration and Authentication			
Login Page/ Sign Up Page	The error messages received when the user enters an invalid input does not give the user a clear idea of the cause of the error. Such initial error messages include: "Firebase: Error (auth/user-not-found)"	Error text for the respective errors were customised, so as to provide users with a more detailed explanation of what caused the error faced	Resolved
	The colour contrast between the text and background makes the text difficult to read	Enhanced our UI by optimising the colour contrast, resulting in clearer and more legible text for an improved user experience	Resolved
Updating Daily Progress			
Daily Log Page	When filling up the input field for "Number of Hours Spent Studying", the phone's keyboard covers up the text input box, making it difficult for users to check the value inputted	-	To be resolved in upcoming weeks
Activity Page	The individual values for each day's activity may be difficult to read from the graph. It would be helpful if users could tap on each point on the line graph, and be shown the day's respective statistics	-	To be resolved in upcoming weeks

LifeQuest

Friends & Battle System

Battle System	The overall game could be more engaging. The implementation seems a little too simple	Enhanced the design of our game by replacing the player avatars, and adding battle animations into the game. A mini-game was also added within the game to increase engagement	Resolved
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7. Timeline and Development Plan

Task	Date	Ryan (hours)	Yogee (hours)	Remarks
Technical knowledge research	8 -15 May	25	25	Familiarisation of the required tech stacks (through online courses, tutorial videos, articles and documentations)
Ideation of poster & video (Liftoff)	10 May	1	1	Conceptualising poster and video drafts
Creation of poster & video (Liftoff)	11 May	0	5	Finalising poster and video
Team meeting & initial planning	13 May	2	2	Determining developmental timeline and deliverables. Ideation of features and system design
Creating app wireframe using Figma	13 May	0	4	Conceptualising the app layout and the flow between the screens
Setting up basic frontend	15 - 25 May	15	0	Implementing login & signup page, Navbar, and Home screens
Setting up basic backend	20 - 25 May	5	0	Setting up GET endpoint to retrieve user's weekly logs & stats from Firestore database
Setting up user authentication	24 - 28 May	4	0	Setup and implement user account creation using Firebase authentication
Submissions for Milestone 1	26 - 28 May	2	5	Finalisation of Milestone 1 submissions (Report, Poster, Video)
Evaluation Milestone 1				

LifeQuest

Improving UI	29 May - 4 Jun	3	0	Adding background images for screens, changing icons used in Navbar & screens, improving readability of alert error messages
Setting up front end for Battle Page	1 - 7 Jun	0	10	Designing and implementation of basic front-end for Battle page
Improving user authentication	5 - 7 Jun	5	0	Implementing “Logout” and “Retrieve password” feature
Linking user stats to Battle Page	7 - 8 Jun	0	5	Retrieving and updating of user and friend stats on the BattlePage
Linking daily log to Firestore database	8 - 12 Jun	6	0	Setting up PUT endpoint to update user’s weekly logs & stats in Firestore database based on “Daily Log” inputs
Setting up backend for Battle Page	10 - 18 Jun	0	10	Setting up the button functions in how they affect health bars and power bars, including the charging feature of the ultimate button.
Setting up friend feature	13 - 20 Jun	20	0	Implementing add/remove friend feature frontend, as well as using GET/POST/DELETE endpoints, such that both user’s Firestore database updates when friend request is sent/revoked
Basic user testing	21 - 23 Jun	10	0	Creating test cases and adding in contingencies to deal with these test cases (Example: Login with invalid email, sending friend request to invalid player ID etc)
Automated Battle System	19 - 23 Jun	0	10	Implementing an automated battle system which follows the rules of the game and ends when a player’s health reaches zero.

LifeQuest

Basic user testing for Battle System	23 - 25 Jun	0	15	Creating test cases and adding contingencies to improve user experience (Example: Disabling buttons to prevent players from disrupting the bot, adding a helper button to read the rules of the game if in doubt)
Submissions for Milestone 2	24 - 29 Jun	6	10	Finalisation of Milestone 2 submissions (Report, Poster, Video)
Evaluation Milestone 2				
Automated Battle System	3 - 10 Jul	12	25	Implementing battle animation, improvement of game design and game logic
Familiarisation with Jest	7 - 10 Jul	4	4	Familiarisation through online courses, tutorial videos, articles and documentations)
Automated Testing	11 - 17 Jul	10	4	Writing of test cases for automated testing and integration testing
User Testing	18 Jul	2	2	User testing using the current iteration of our mobile application, and gathering feedback from our friends (who serve as our target audience)
Improving UI and Minor Adjustments	19 - 20 Jul	2	2	Making minor changes to the UI, Battle System and Error Alerts based on previous user feedback
Submissions for Milestone 3	21 - 23 Jul	8	8	Finalisation of Milestone 3 submissions (Report, Poster, Video)
Evaluation Milestone 3	24 Jul	142	147	Total Hours Spent